

Reaction extent before routes to K

Read and prepare: 15–20 minutes total. Bring tentative answers; uncertainty is useful for discussion.

How does a reaction choose its equilibrium composition?

Extent is the composition coordinate

Write $n_i = n_{i,0} + \nu_i \xi$, with negative stoichiometric coefficients for reactants. Nonnegative species amounts determine the feasible extent interval. Before calculating K, identify the reaction basis, conserved atoms and standard states. The same extent value has different meanings for differently scaled reactions.

Driving force along the feasible path

At fixed T and P, $dG/d\xi = \Delta_r G = \sum \nu_i \mu_i$. For an interior equilibrium with the required species present, $\Delta_r G = 0$. Using dimensionless activities gives $\Delta_r G = \Delta_r G^\circ + RT \ln Q$. Thus $Q = K$ at equilibrium and $\ln K = -\Delta_r G^\circ / (RT)$. A boundary minimum needs a one-sided condition; never fabricate a zero-amount logarithm.

K comes from thermodynamic information

Once the equilibrium criterion is clear, obtain $\Delta_r G^\circ$ from direct data, formation properties, or $\Delta_r H^\circ - T\Delta_r S^\circ$ at the evaluation temperature. Reconcile units and reaction basis before comparing routes. Varying pressure at fixed T changes ideal-gas activities and may change extent, while K stays fixed for the same reaction and standard states. Temperature variation requires applicable thermal data.

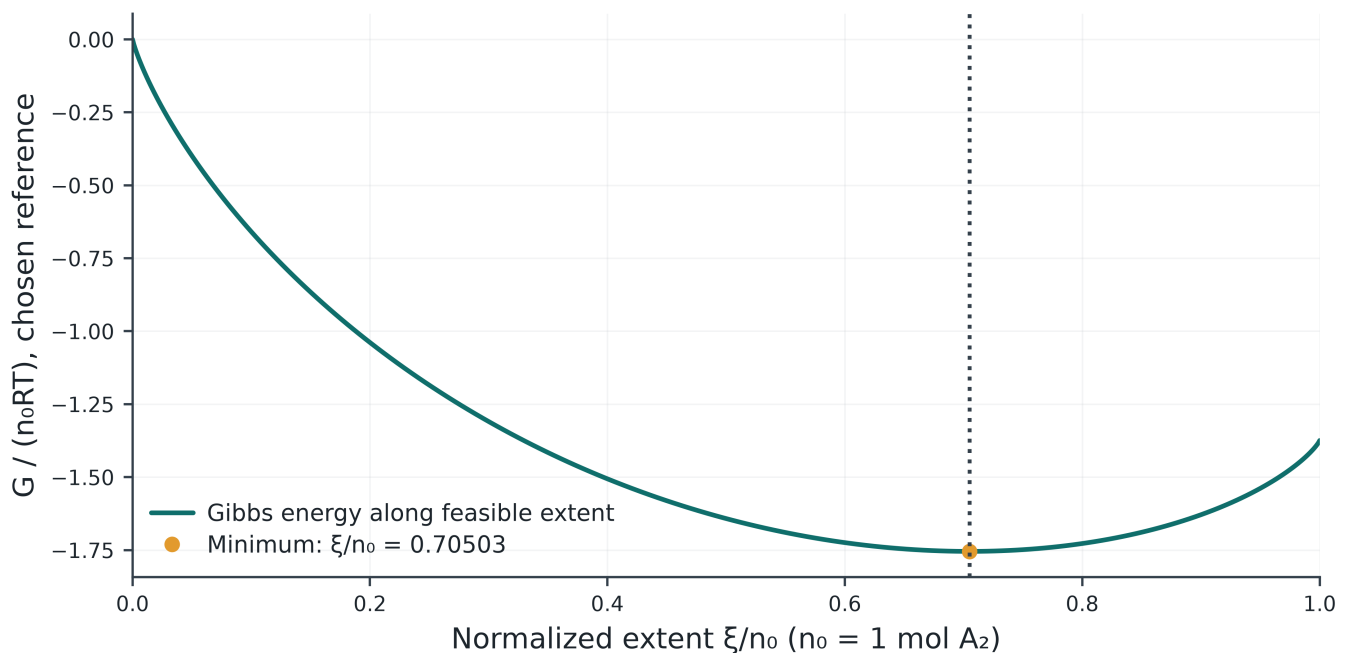
$$n_i = n_{i,0} + \nu_i \xi; \quad dG/d\xi = \Delta_r G^\circ + RT \ln Q$$

$$Q = \prod a_i^{\nu_i}; \quad \ln K = -\Delta_r G^\circ / (RT)$$

One example to reason through

Synthetic teaching values. These numbers illustrate the assumptions; they are not measurements of a real material.

Use the Lab 12 synthetic reaction $A_2 \rightleftharpoons 2A$ at 350 K, with one initial mole of A_2 , no A and no inert. Then $n_{A_2}=1-\xi$ and $n_A=2\xi$, so $0 \leq \xi \leq 1$ mol. At $P=P^\circ=100$ kPa, the numerical extent in moles gives $Q=4\xi^2/(1-\xi^2)$. With $\Delta_r G^\circ=-4000$ J/mol, $\ln K=1.37454$ and $K=3.95326$. The interior minimum occurs at $\xi \approx 0.70503$ mol. Both the Gibbs-energy minimum and $Q=K$ identify the same composition.



Check and interpret

At $\xi=0.5$ mol, $Q=1.33333$, below K , so $\Delta_r G$ is negative and increasing extent lowers G . At $\xi=0.9$ mol, $Q \approx 17.0526$, above K , so the reverse direction lowers G . Doubling pressure at fixed T increases Q at a given extent for this reaction, so equilibrium shifts toward lower extent.

Bring three short answers

A sketch, a few numbers or one sentence is enough. Do this before opening the calculator; revise your prediction in class.

1. Write species amounts and the feasible extent interval before using any K formula.

2. At $\xi=0.5$ mol in the example, predict the favorable direction from Q/K.

3. Why does the pressure effect depend on stoichiometry, while K at fixed T does not?

Optional extension

For multiple reactions, use independent reaction coordinates or minimize Gibbs energy with atom balances. This is reference material beyond the single neutral ideal-gas extent solver. A pure solid may have unit activity while present, but disappearance and reaction feasibility still require checks.

Continue in the core lab

<https://www.skhgroup.net/teaching/2105603/labs/reaction.html>

Reading locator: supplied textbook contents, sections 17.1-17.3; optional 17.5 and 17.8.

Original course explanation and examples, not a reproduction of the textbook. Section numbers are based on the supplied contents pages. Required weekly independent work: 15–20 minutes preparation plus 30–40 minutes follow-up; optional extensions are additional.